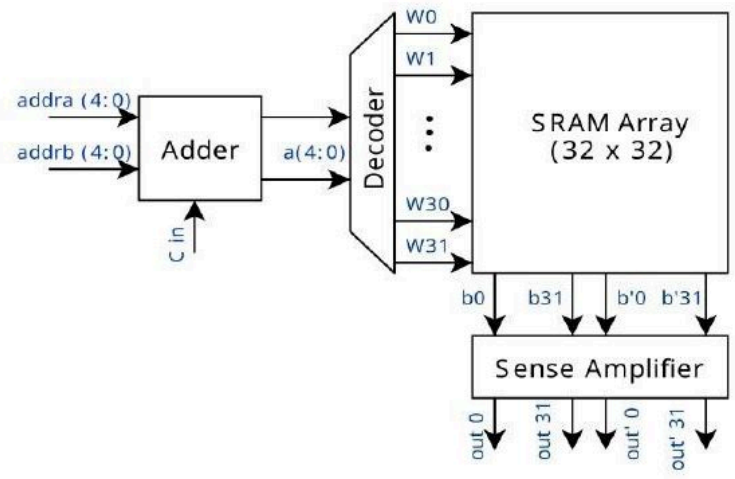
Design and Analysis of a 32x32-bit SRAM

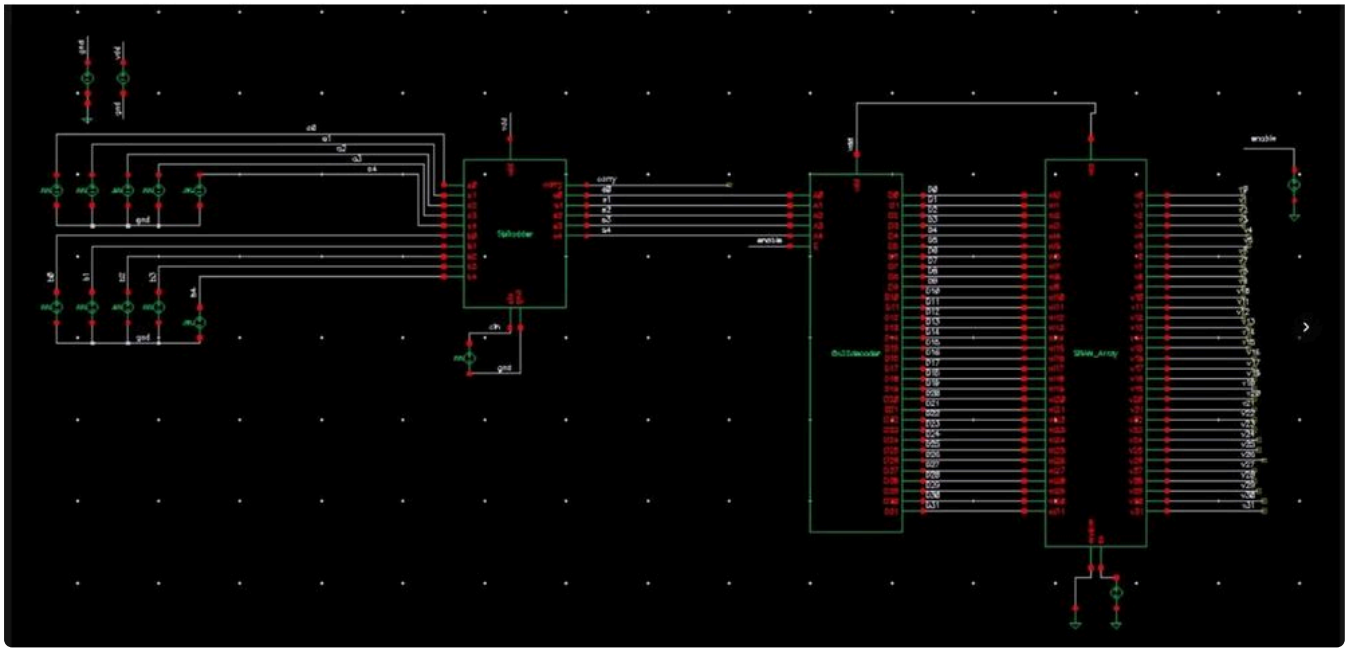
TSMC 28 nm. Team: Pragnya (2023102067), Aikya (2023102062), Indira (2023102070), Abhinaya (2023102071).

1. Introduction

This project covers a 32 x 32-bit SRAM in the TSMC 28 nm process, built from the 6T bit cell up to a full read path. Words are addressed with a 5-bit address so the memory can be used as a FIFO. Cells are pre-loaded through initial conditions, so only read operation is simulated.

- **Adder**: adds two 5-bit addresses (addra, addrb) and a carry-in to give the 5-bit word address.
- **Decoder**: converts the 5-bit address to one-hot wordlines W0 to W31, gated by an active-high enable.
- **SRAM array**: 32 words of 32 cells (1024 cells). The active wordline puts its row on the bitline pairs BL/BLB.
- **Sense amplifier**: resolves the small bitline difference into a full-swing output, one per column.





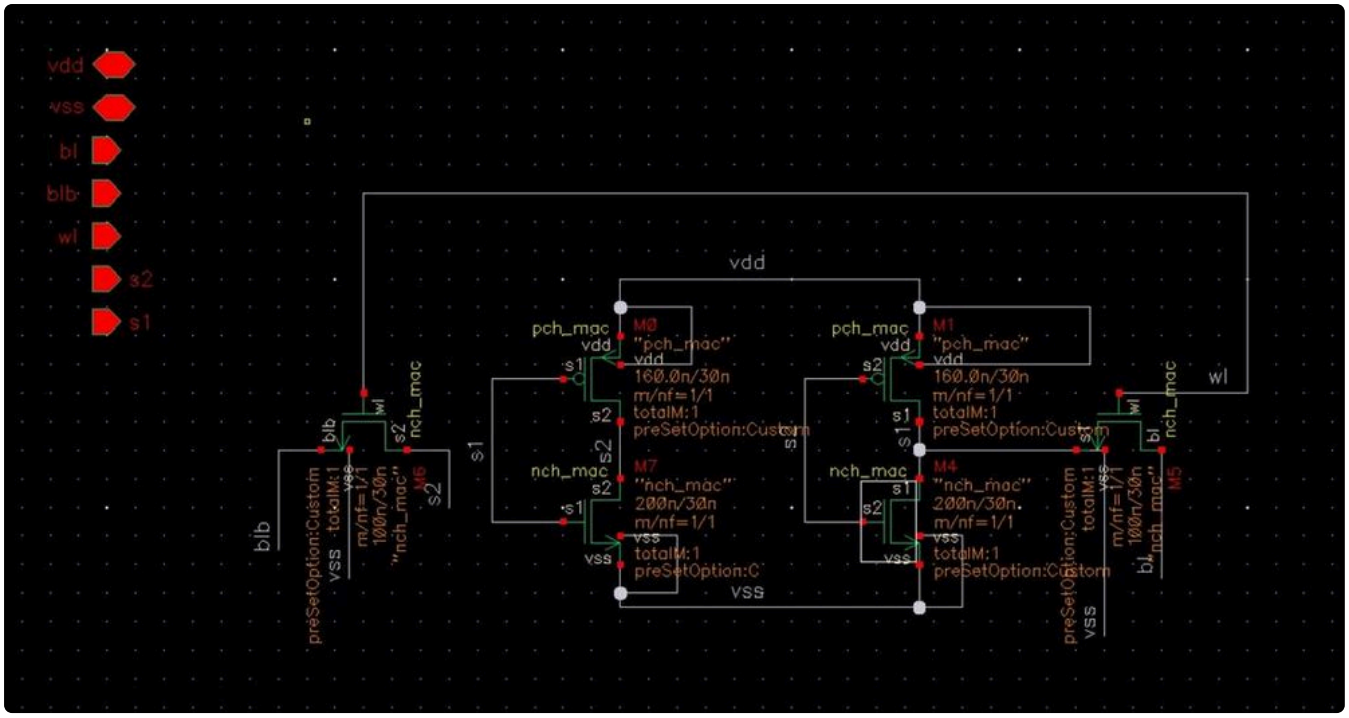
On a read, the adder output is decoded, the selected wordline turns on the access transistors of one row, one bitline of each pair discharges slightly, and the sense amplifier regenerates that difference into a logic level.

1.1 Simulation setup

All circuits were simulated in Cadence Virtuoso with the TSMC 28 nm models at a nominal supply of 1.0 V. Every block has its own testbench, and delays are measured between 50% crossings of the input and output.

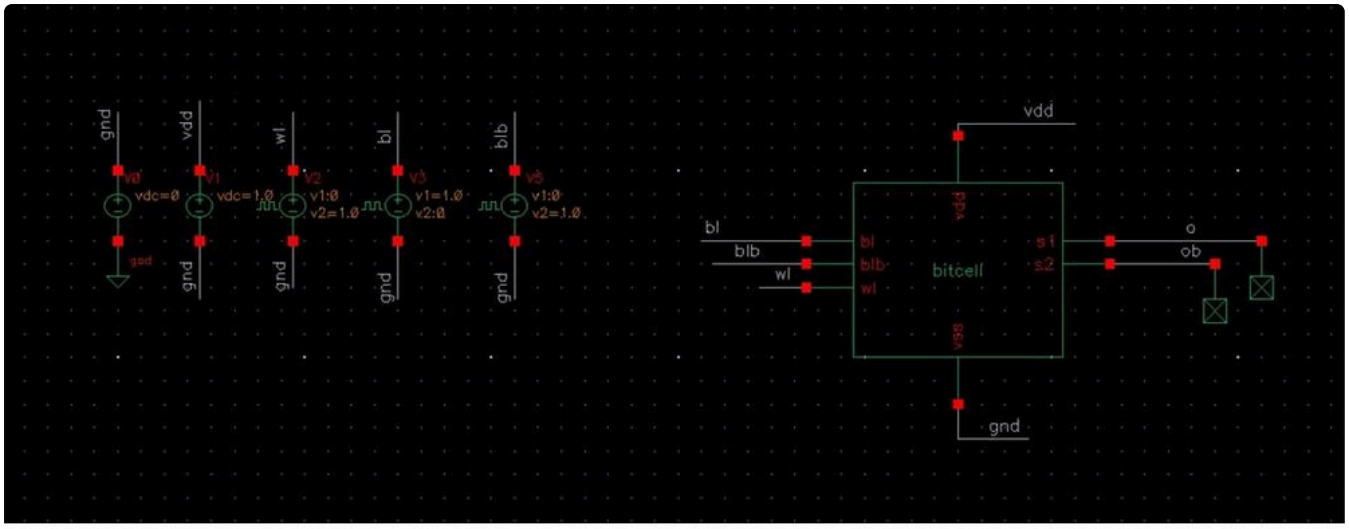
2. 6T SRAM Cell

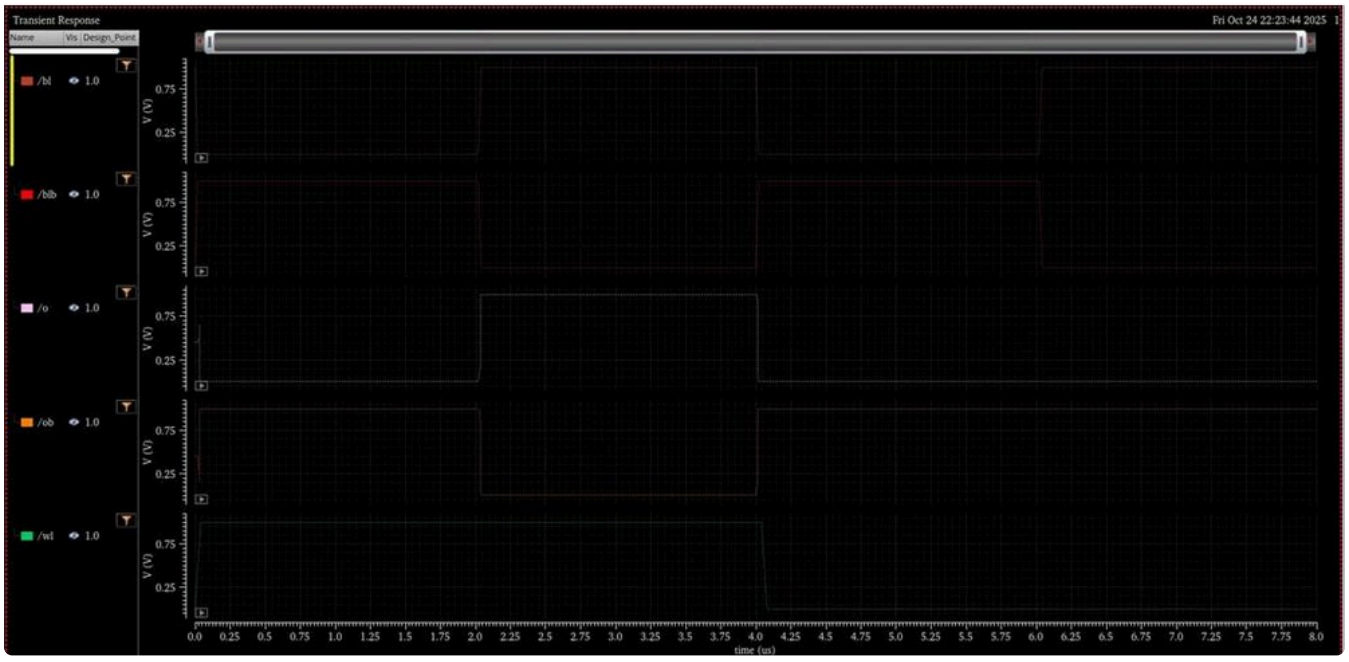
The cell is two cross-coupled inverters (a pull-up PMOS and a pull-down NMOS each) plus two NMOS access transistors driven by the wordline. The positive feedback gives two stable states, Q = 1 / Q′ = 0 and Q = 0 / Q′ = 1, so the bit is held as long as VDD is applied.



2.1 Operation

- **Hold (WL = 0)**: access transistors are off, the cell is isolated from the bitlines and keeps its state.
- **Read (WL = 1)**: BL and BLB are precharged to VDD. The node storing 0 pulls its bitline down slightly through the access and pull-down transistors, while the other bitline stays high. The sense amplifier resolves the difference.
- **Write (WL = 1)**: one bitline is driven to 0 V and the other to VDD. The access transistor on the low side overpowers the pull-up PMOS and the feedback completes the flip.





2.2 Transistor sizing

Device	Purpose	W / L
Pull-down NMOS	holds the stored 0; must out-drive the access device on a read	200 nm / 30 nm
Pull-up PMOS	holds the stored 1; kept weaker so a write can overpower it	160 nm / 30 nm
Access NMOS	connects the node to its bitline; small to limit read disturb, strong enough to write	100 nm / 30 nm

- **Read stability**: the node at 0 rises when the access transistor starts to discharge the bitline, so the pull-down must be stronger than the access device. The cell ratio sets how far the node rises.

$CR = (W/L)_{pd} / (W/L)_{access} = 200 / 100 = 2$

- **Writability**: the access transistor must overpower the pull-up. The PMOS is already weaker per unit width, and it is kept narrower than the pull-down.

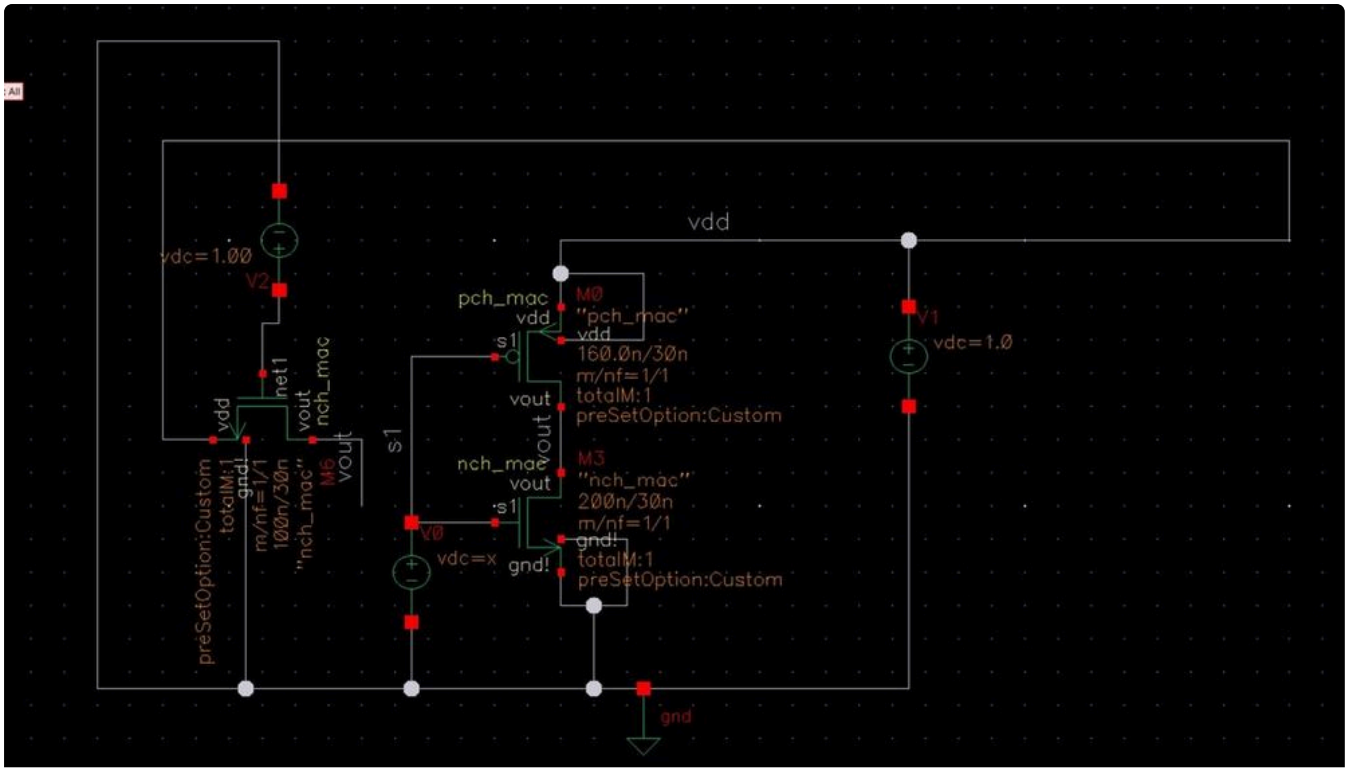
$I_{access} / I_{pu} \approx (\mu_n / \mu_p) \times (W/L)_{access} / (W/L)_{pu} \approx 2.5 \times 100 / 160 \approx 1.6$

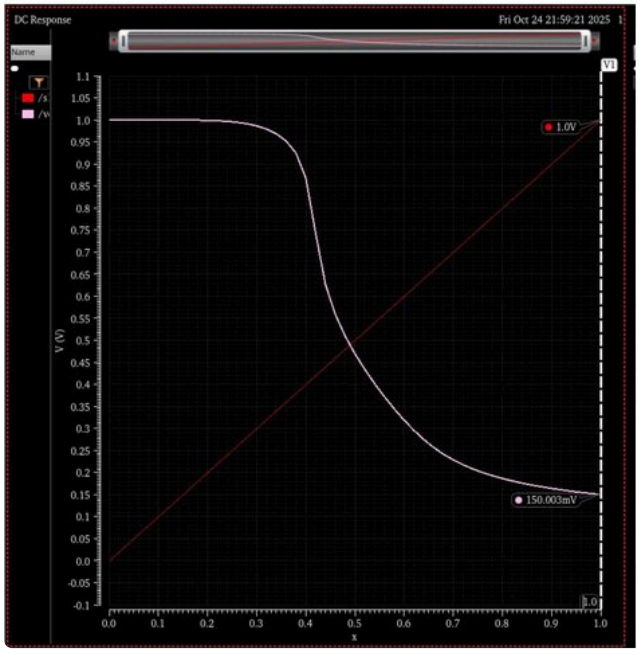
- **First-order read SNM** for VDD = 1 V:

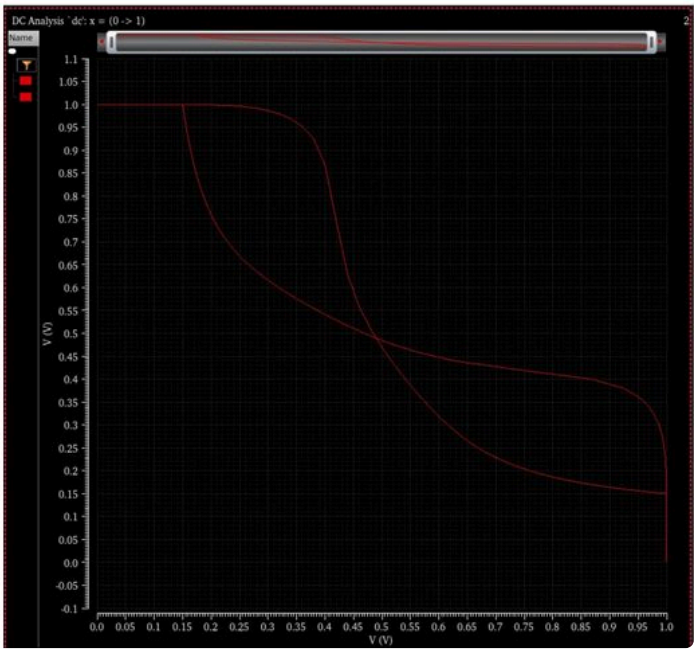
$SNM \approx (VDD / 2) \times (CR - 1) / (CR + 1) = 0.5 \times 1/3 \approx 0.17\ V$

2.3 Read static noise margin

The feedback is broken and one half of the cell is swept: V1 (input) from 0 to VDD with the access transistor on and the bitline at VDD, measuring V2. Mirroring the VTC about V1 = V2 gives the butterfly plot, and the read SNM is the side of the largest square that fits inside it.



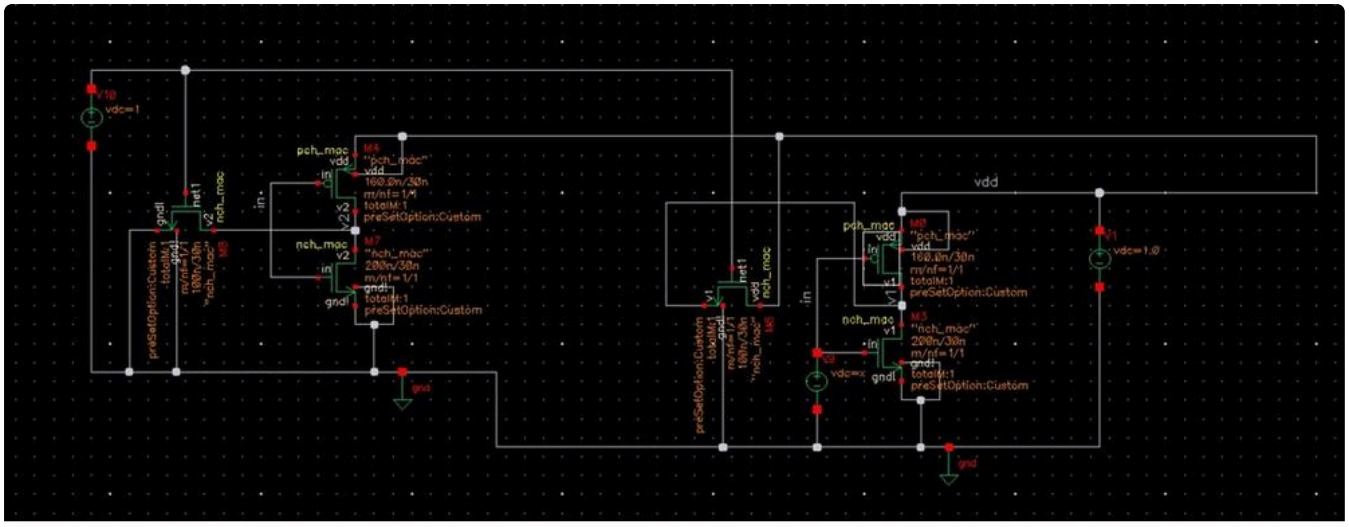


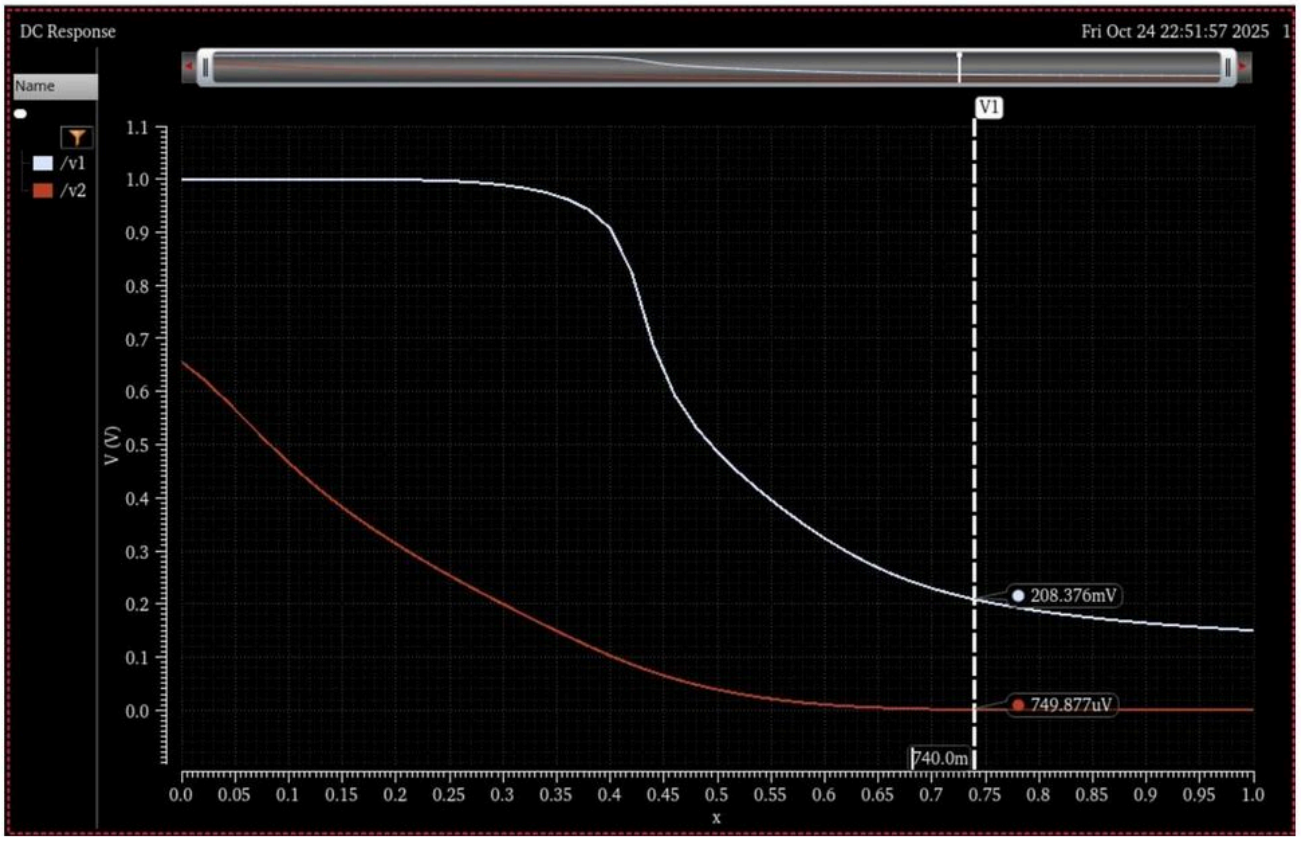


- Both curves cross close to VDD / 2, so the two inverters are well matched.
- Worst-case voltage rise at the 0 node during a read (V2 at V1 = VDD): **150.003 mV**, well below the inverter switching threshold, so the cell is not disturbed.

2.4 Write noise margin

For the write, one bitline is held at 0 V and the other at VDD, so the two half-cell VTCs are no longer identical. Plotting them against each other gives the write butterfly, and the WNM is the side of the largest square between the curves.



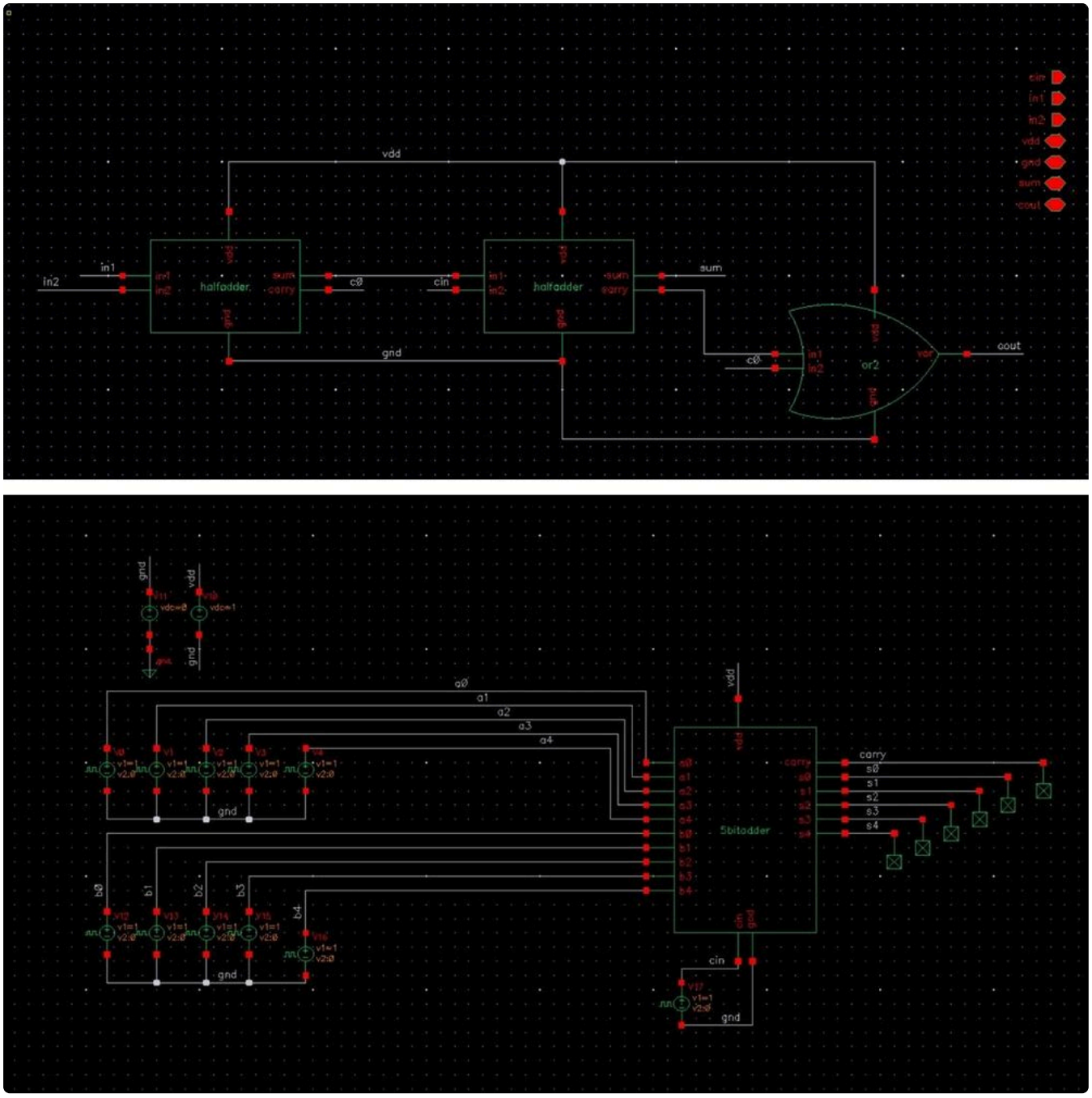


- Worst-case cell voltage during a write (V1 with V2 at 0 V): **208.376 mV**.
- The 2 : 1 cell ratio protects the read, and the access to pull-up ratio of about 1.6 keeps the cell writable. Making the access transistor stronger would help the write but reduce read stability.

3. Adder

The baseline adder is a 5-bit ripple-carry adder: five full adders in static CMOS, each made of two half adders and an OR gate.

$S_i = A_i \oplus B_i \oplus C_{in}$, $C_{out} = A_i \cdot B_i + C_{in} \cdot (A_i \oplus B_i)$

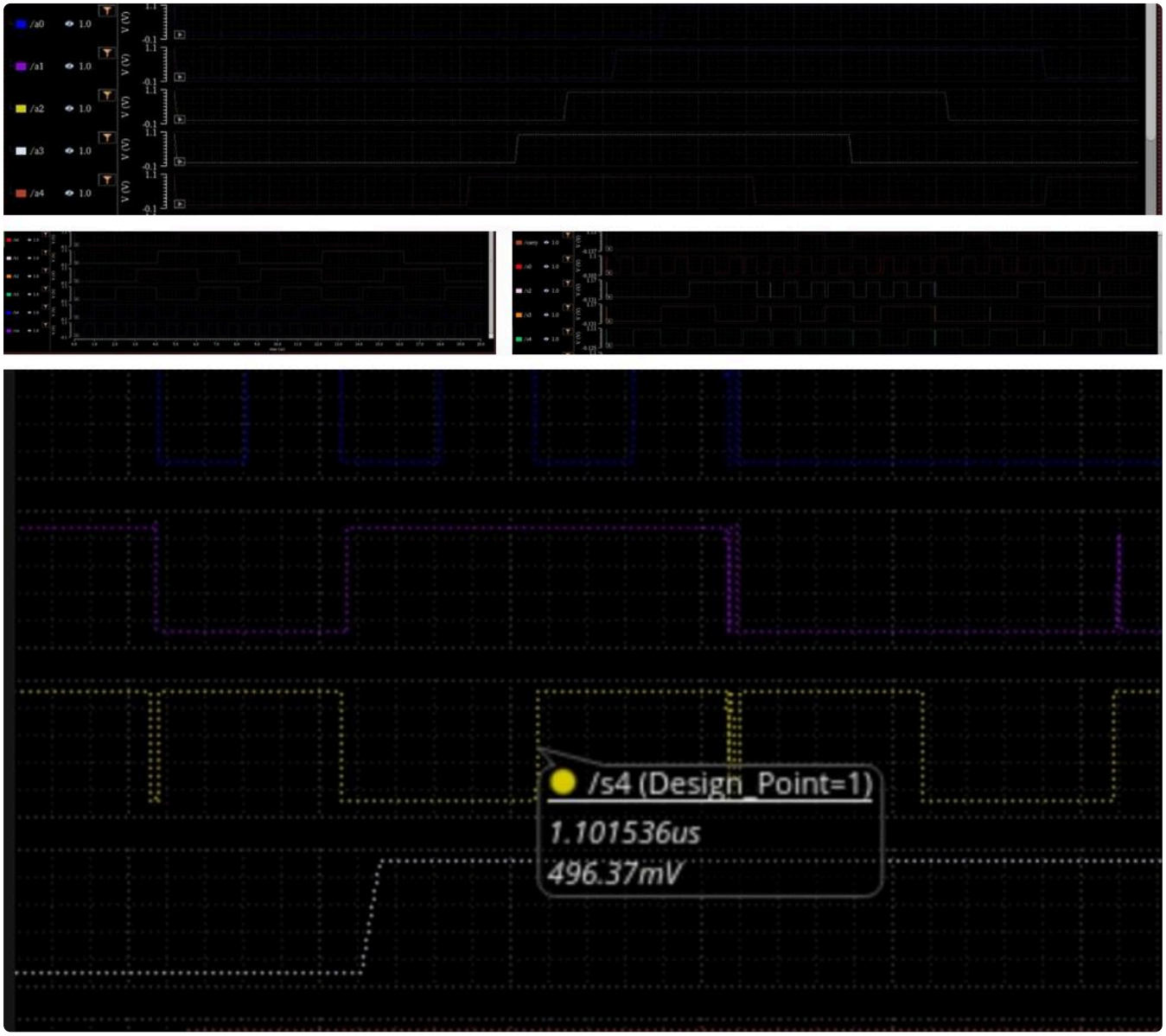


3.1 Critical path

The sum of the last bit is valid only after the carry has rippled through every stage, so the worst case is a full-length carry chain: A = 11111, B = 00001, Cin = 0. Each stage adds about two gate delays (AND, then OR) on the carry path, and the final sum adds an XOR.

Cin → C0 → C1 → C2 → C3 → C4 (S4)

3.2 Simulation



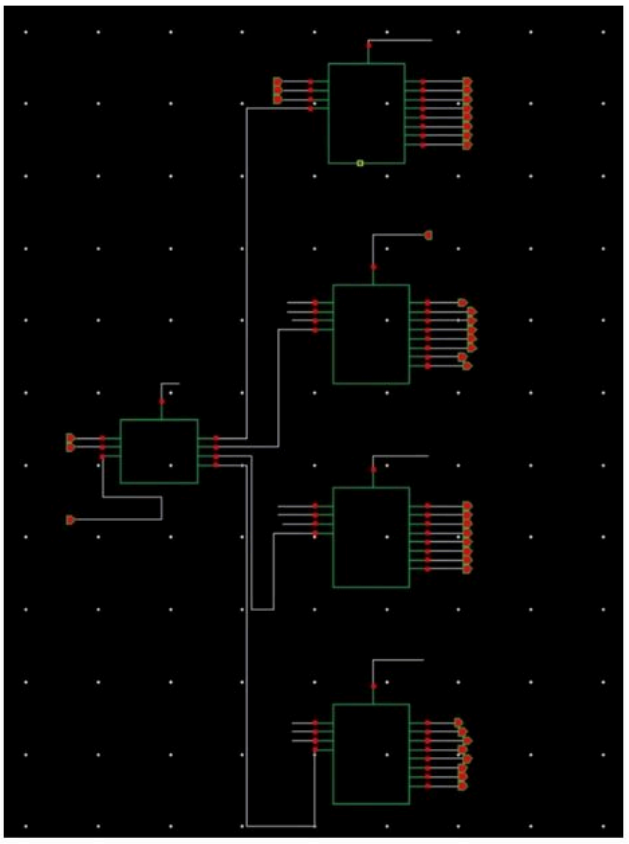
- Estimated worst-case delay: **about 110 ps**, from roughly 11 gate delays on the carry-to-sum path at about 10 ps each. The per-gate figure comes from the decoder simulation (eight stages in about 100 ps, or 12 ps per stage), a little lower here because the carry gates drive a lighter load.
- The delay grows linearly with the number of bits. This is the main drawback of the ripple structure and the reason the adder is revisited in Section 8.

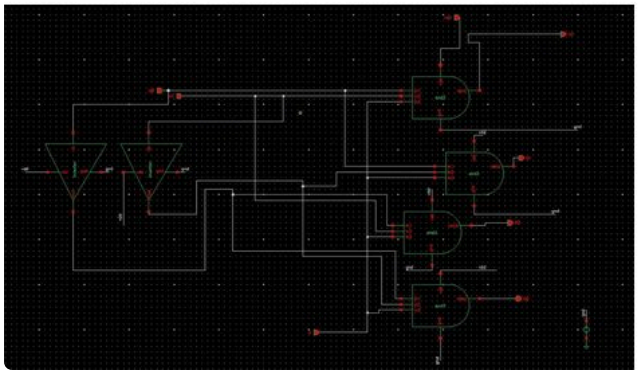
4. Decoder

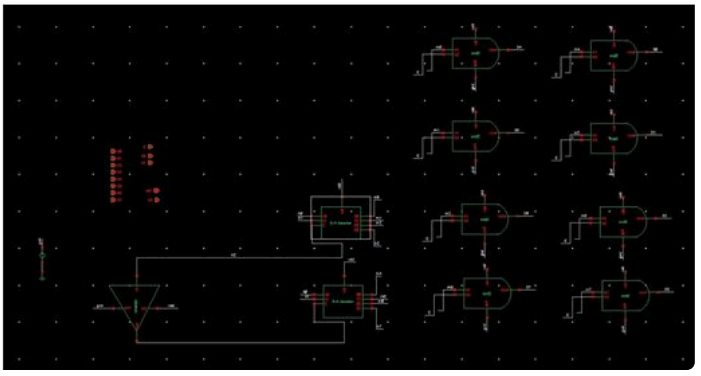
The 5-to-32 decoder drives a wordline loaded by 32 access-transistor gates, so electrical and branching effort are high and the design is sized with logical effort. A single 5-input gate per output would be slow, so the decoder is split hierarchically and built only from NAND2 gates and inverters (NAND2 has a lower logical effort than NOR2: 4/3 against 5/3).

4.1 Architecture

- A **2-to-4 pre-decoder** on A[1:0] produces four enable signals.
- Each enable drives one of four **3-to-8 decoders** on A[4:2], so 4 x 8 = 32 wordlines.
- Each output is gated by the active-high **EN** signal. With EN = 0 every wordline stays low, which lets the bitlines precharge between reads.







4.2 Logical effort

The critical path is address input, inverter, NAND2, inverter, NAND2, inverter, NAND2, inverter, wordline: N = 8 stages with three NAND2 gates.

Parameter	Value
Path logical effort G = (4/3)^3	2.37

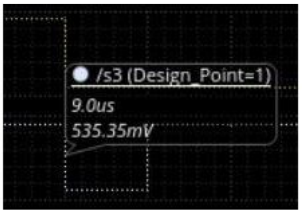
Parameter	Value
Electrical effort H (wordline load / input)	50
Branching effort B	2
Path effort F = G B H	237
Stage effort f = F^(1/N)	1.98
Parasitic delay P (3 NAND2 + 5 INV)	11

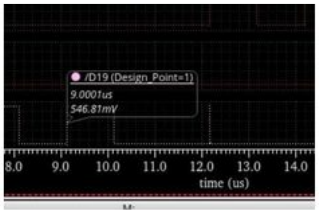
D = N·f + P = 8 × 1.98 + 11 ≈ 27 τ

Every stage is sized backwards from the wordline load so that each carries the same effort f.

4.3 Simulation

The measured delay is from the address input crossing to the selected wordline rising.

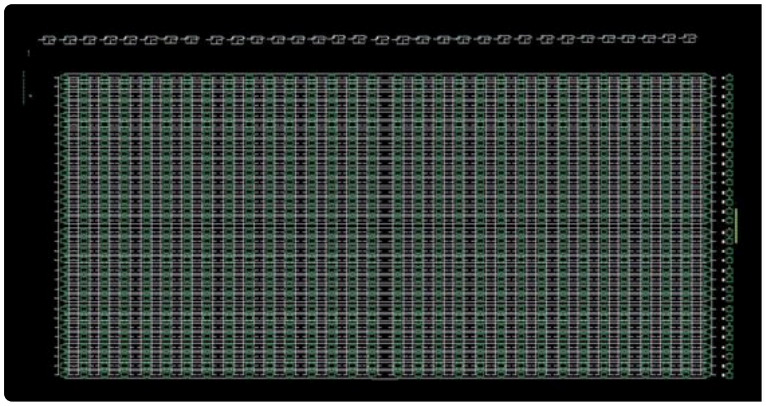


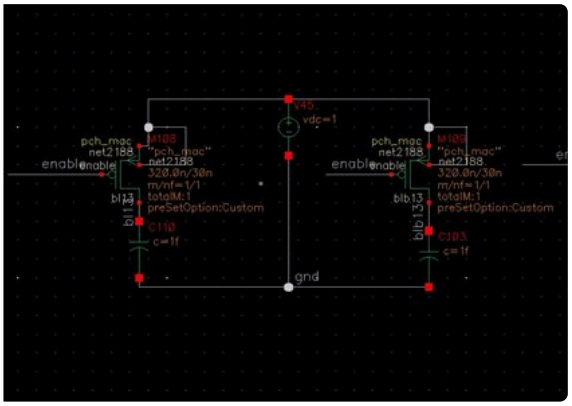


- Decoder delay: **about 100 ps**.

5. SRAM Array and Precharge

The array is 32 rows (wordlines WL0 to WL31) by 32 columns (bitline pairs BL0/BLB0 to BL31/BLB31), 1024 6T cells in total. A wordline connects all 32 cells of a row to their bitlines at once, and each bitline loads 32 access transistors.





5.1 Precharge circuit

Before every read, both bitlines of each column are charged to VDD so the sense amplifier sees a known starting point and a small difference can develop.

- One PMOS (320 nm / 30 nm) per bitline, 64 in total, driven by an active-low enable.
- When the enable is low the PMOS devices are on and the bitlines charge to VDD. The enable is released before the wordline rises.
- Constraint: the bitlines must recover to within 10% of VDD faster than the adder and decoder delay (about 210 ps for the baseline adder and decoder, about 160 ps with the Manchester adder), so the next read is not delayed.

5.2 Bitline development

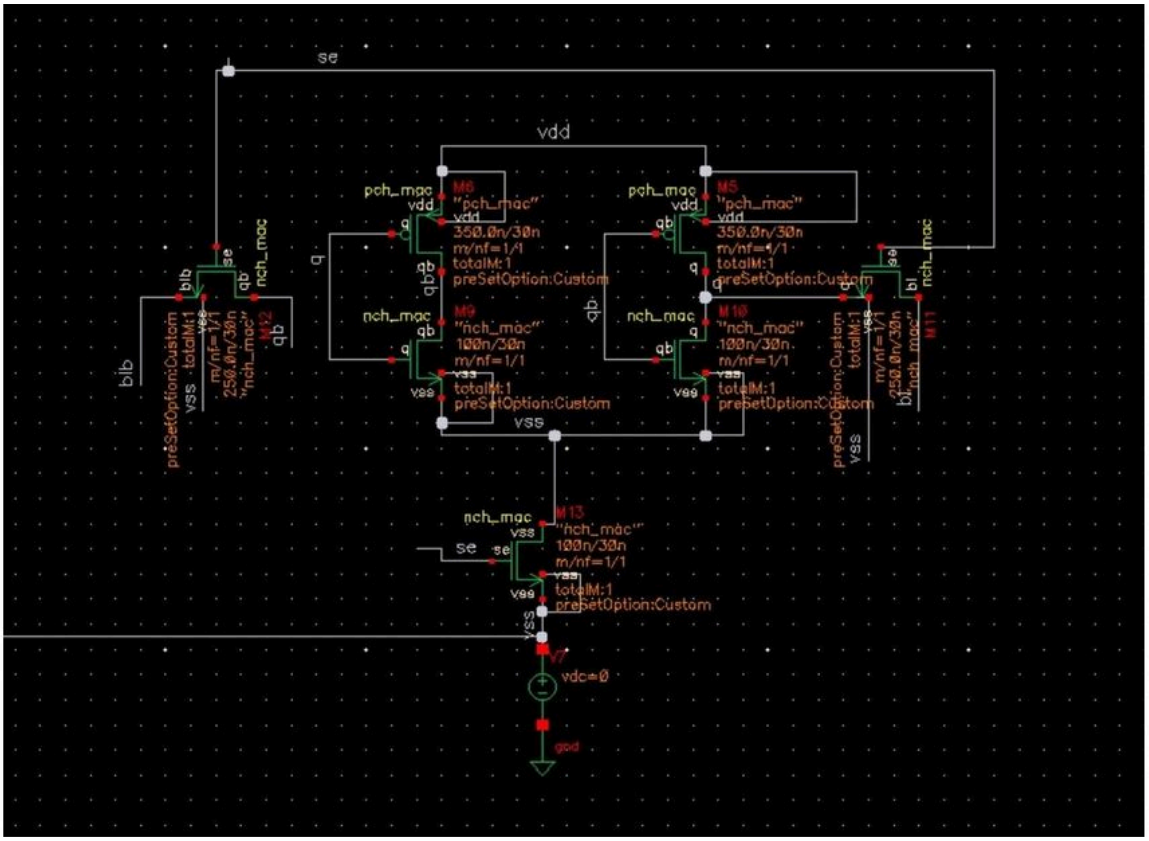
During a read the selected cell discharges one bitline through its access and pull-down transistors. The sense amplifier is enabled once the difference reaches about 100 mV.

$t_{BL} = C_{BL} \cdot \Delta V / I_{read} \approx (10\ \text{fF})(0.1\ \text{V}) / (33\ \mu\text{A}) \approx 30\ \text{ps}$

- C_{BL} of about 10 fF: 32 access-transistor drains, the bitline wire, the precharge PMOS and the sense amplifier input.
- I_{read} of about 33 uA: the access and pull-down NMOS in series at 1.0 V.
- The wordline has to stay on at least this long, which sets the sense-enable timing.

6. Sense Amplifier

A read leaves only a small difference between BL and BLB, so each column has a sense amplifier that amplifies it to a full-swing output. It is a cross-coupled latch amplifier: positive feedback regenerates the small input difference, which is much faster than waiting for the bitline to discharge fully.

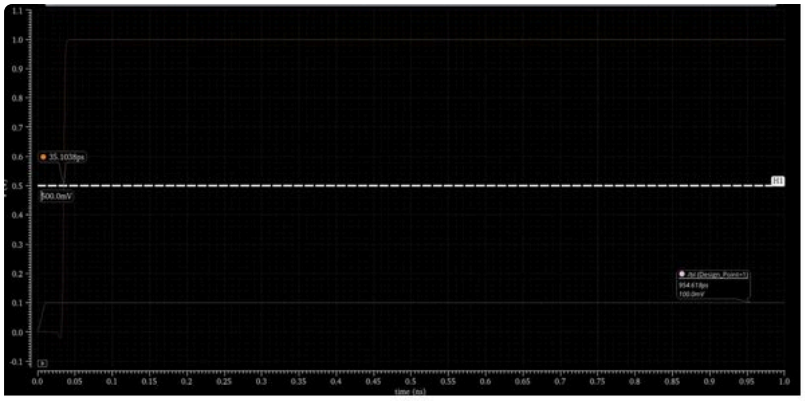


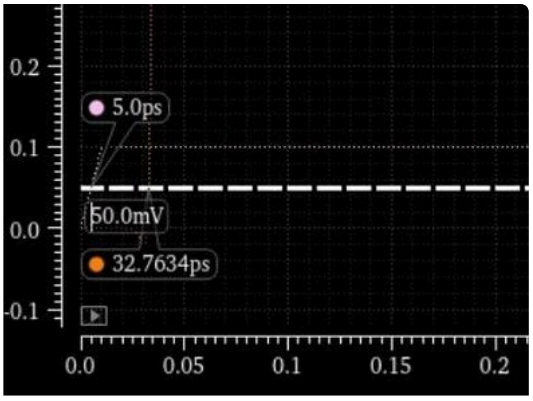
6.1 Circuit and operation

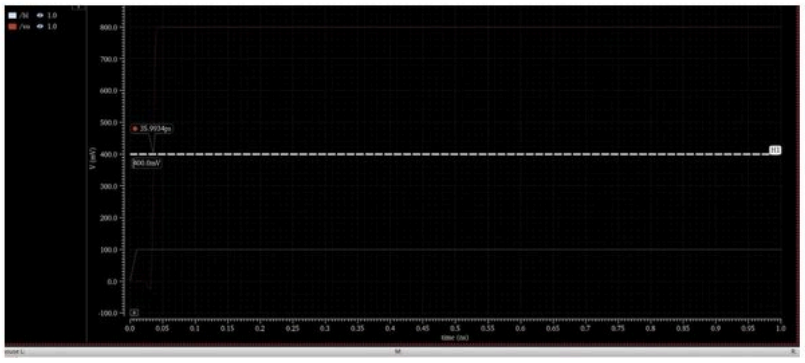
- **Latch**: PMOS 350 nm / 30 nm and NMOS 100 nm / 30 nm in two cross-coupled inverters, with outputs q and qb.
- **Input devices**: NMOS 250 nm / 30 nm (M11, M12), gated by SE, couple BL and BLB onto the latch nodes.
- **Tail**: NMOS 100 nm / 30 nm gated by SE. The amplifier is idle until SE rises, which saves power.
- With SE high, the side connected to the lower bitline is pulled down, the feedback flips the latch and q/qb settle at VDD and 0. A stored 0 (BL lower) gives output 0 and a stored 1 (BLB lower) gives output 1.

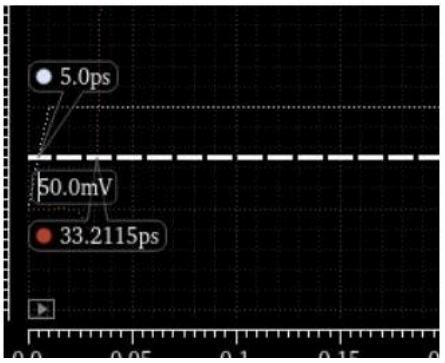
6.2 Delay

The delay is measured from the bitline input crossing 50 mV (5.0 ps) to the output crossing 50% of its swing, with a 100 mV bitline difference applied.









Supply	Input crossing	Output at 50% swing	Delay
1.0 V	5.0 ps	35.10 ps	30.1 ps
0.8 V	5.0 ps	35.99 ps	31.0 ps

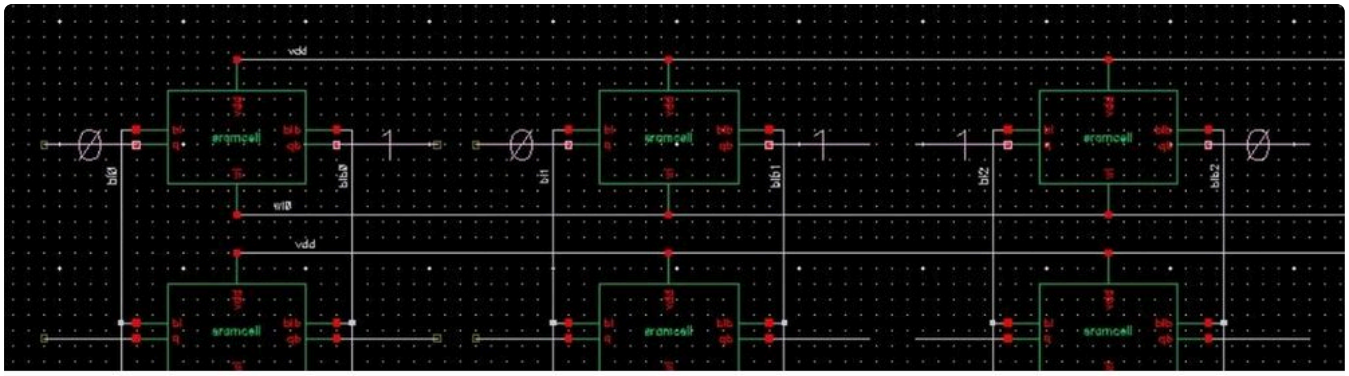
The lower two plots show the 0.8 V case, which is used in Section 8.2.

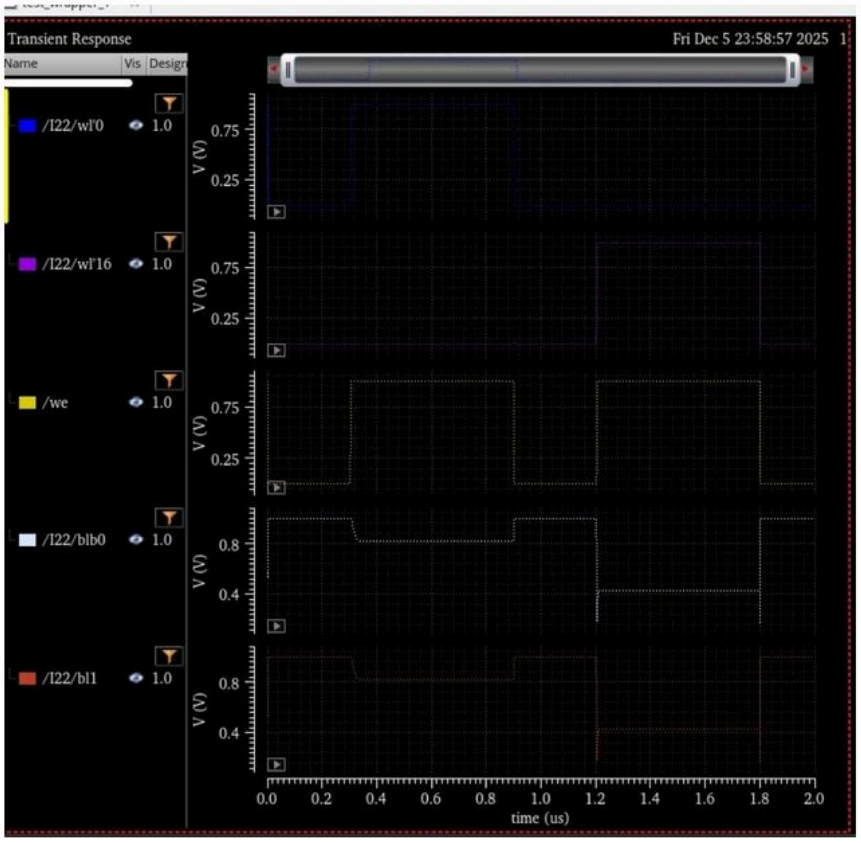
7. Complete System

The four blocks are connected as in the block diagram of Section 1 and simulated from the adder inputs to the sense amplifier outputs.

7.1 Back-to-back reads

Two reads are performed on two different rows (WL0 and WL16), with the stored data initialised so that the output flips polarity between the two cycles.



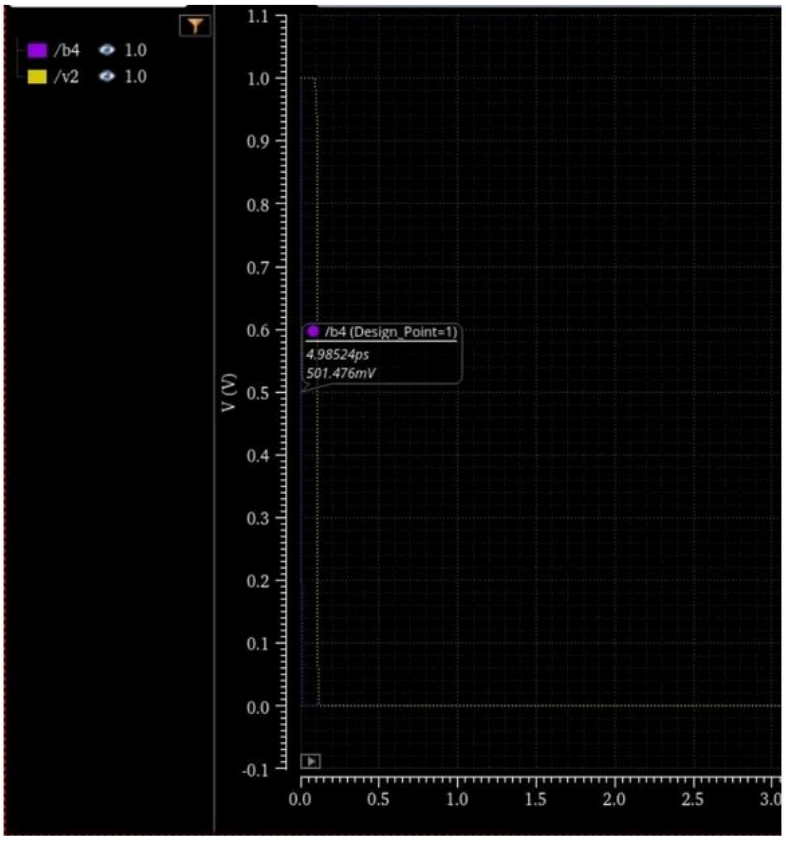


- WL0 rises, the selected bitline of the column falls and the sense amplifier resolves the stored value.
- WL0 is fully off before the bitlines return to VDD through the precharge devices.
- WL16 then rises and the second read repeats the sequence with the opposite data. The first read does not corrupt the second.
- The bitlines fall from 1.0 V to about 0.82 V in the first read and to about 0.43 V in the second, because the wordline is held for 600 ns, far longer than the sense amplifier needs. The sense amplifier output swings rail to rail, 0 to VDD (0 to 0.8 V in the optimized version).

7.2 Critical path

address inputs → adder → decoder → wordline → bitline → sense amplifier → output

The latency is measured from the transition at the adder input to the sense amplifier output crossing 50% of VDD.



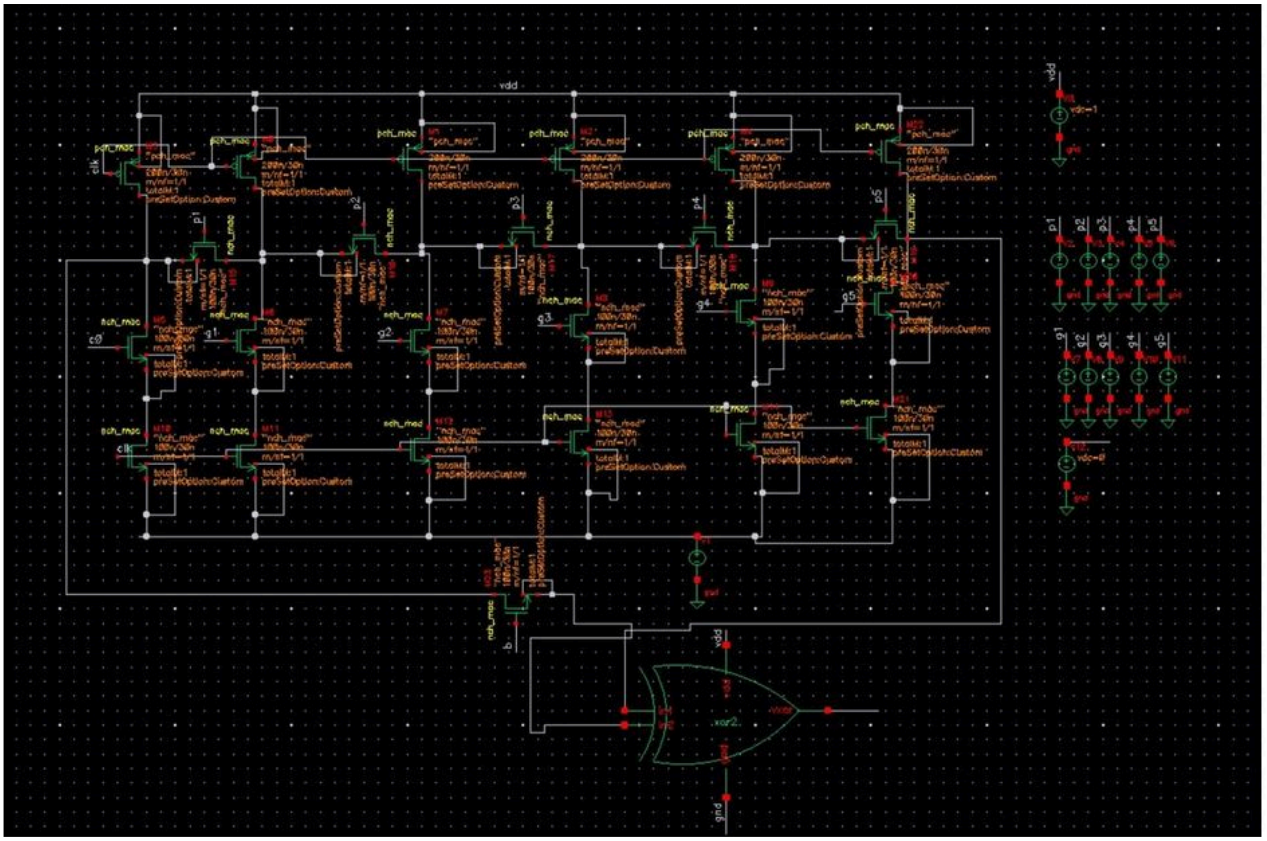
- The worst case combines the full carry chain in the adder (Section 3.1), an address change that switches all eight decoder stages, a bitline pair with the smallest difference, and the lowest supply.
- In the baseline the adder (about 40% of the latency) and the decoder (about 37%) dominate. After the adder optimization the decoder is the largest part, about 45%. Block delays are listed in Section 9.

8. Optimizations

The baseline design was first made functional. The changes below were then applied to cut latency and power.

8.1 Adder: Manchester carry chain

The ripple-carry adder was replaced by a Manchester carry chain. Propagate (P) and generate (G) signals are formed per bit and the carry runs through a single chain, with an XOR sum stage, instead of through two gate delays per bit.



- Fewer series gates on the carry path, with the critical-path devices upsized for drive.
- Estimated worst-case delay falls from about **110 ps to about 60 ps**: propagate/generate about 10 ps, carry chain (restored every two bits) about 30 ps, XOR sum about 20 ps.
- Trade-off: pass-transistor stages degrade logic levels, so the chain needs restoring stages, and the circuit is more complex than the ripple adder.

8.2 Supply voltage partitioning

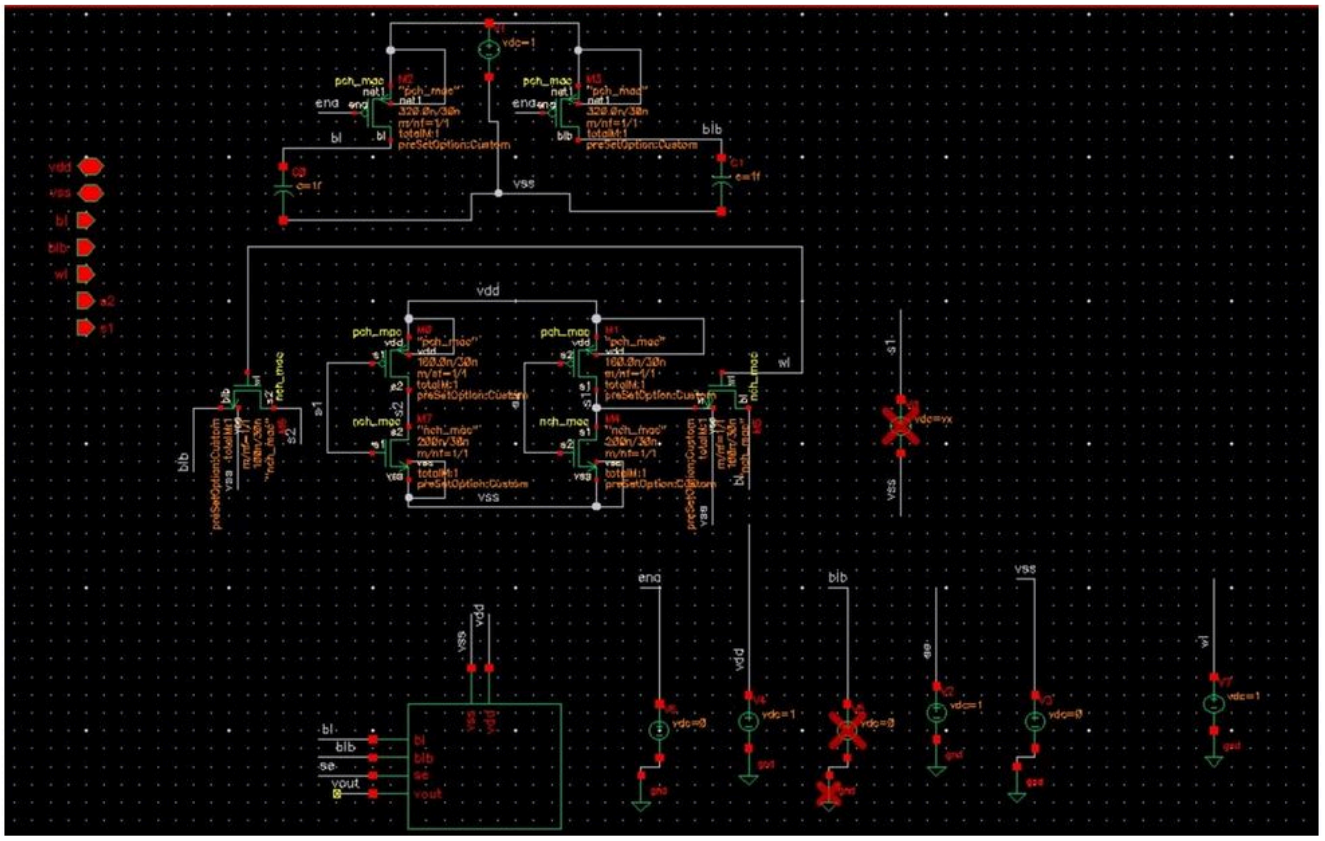
The sense amplifier output swing was reduced by running it from 0.8 V while the cell array stays at 1.0 V. A smaller swing charges and discharges less capacitance every cycle. The simulated waveforms are the lower two plots in Section 6.2.

- Sense amplifier delay stays at about 31 ps (30.1 ps at 1.0 V), so the saving costs almost no speed.
- Switching energy scales with VDD squared, so (0.8 / 1.0)^2 = 0.64: about **36% lower** for the sense amplifier.
- Trade-off: two supply rails, more routing, and a reduced output swing for the stage that follows.

8.3 Precharge and wordline drivers

- **Precharge**: the PMOS devices (320 nm / 30 nm) are sized to bring the bitlines back to VDD in about 50 ps, inside the adder plus decoder delay.

- **Wordline drivers**: drivers on the decoder outputs make sure two wordlines are never on at the same time, so a second row cannot corrupt the data being read.
- Trade-off: larger precharge devices add bitline capacitance and area.



8.4 Output buffers

Each sense amplifier output drives a two-stage inverter chain with increasing sizes. This isolates the latch from the external load so the load cannot slow or disturb the regeneration. The buffer width matches the column pitch, and it lies after the sense amplifier output, outside the latency path as defined in Section 7.2.

9. Latency, Power and Comparison

9.1 Delay

Block	Basis	Baseline	Optimized
5-bit adder	gate-level estimate	about 110 ps	about 60 ps
5-to-32 decoder	simulation	about 100 ps	about 100 ps
Bitline development	C ΔV / I	about 30 ps	about 30 ps
Sense amplifier	simulation	30.1 ps	31.0 ps (0.8 V supply)

Block	Basis	Baseline	Optimized
End-to-end read latency	sum of the above	**about 270 ps**	**about 220 ps**

The adder optimization reduces the read latency by about 18%. In the baseline the adder is the largest block (about 41%); after optimization the decoder is the largest (about 45%). Further gains would come from a faster decoder style such as domino logic.

9.2 Power

Dynamic power follows P = alpha C VDD^2 f, so lowering the supply on the sense amplifier cuts its switching power quadratically while its delay is nearly unchanged.

- Precharge PMOS current: 25.22 uA, about 25 uW.
- Transient supply power in the read simulation: 1 V x 450 uA = 0.45 mW (average over the read cycle).
- Sense amplifier switching energy drops by about 36% at 0.8 V, with no change in delay.
- No static power path when idle: the sense amplifier tail is gated by SE and all wordlines are off, so the cells only hold their state.

9.3 Summary of changes

Block	Baseline	Optimization	Benefit	Cost
Adder	Ripple carry	Manchester carry chain	about 110 to 60 ps	restoring stages, more complex
Sense amplifier	1.0 V supply	0.8 V supply	36% lower switching energy at equal delay	second supply rail
Precharge	PMOS per bitline	320 nm PMOS, about 50 ps recovery	next read not delayed	extra bitline load, area
Wordlines	direct decoder outputs	wordline drivers	no overlapping rows	extra stage, area

10. Conclusion

A 32 x 32-bit SRAM was designed in TSMC 28 nm from the 6T cell up to the sense amplifier outputs, and two back-to-back reads from different rows return the correct, opposite data. The 2 : 1 pull-down to access ratio limits the read disturb at the 0 node to 150 mV, and an access to pull-up ratio of about 1.6 keeps the cell writable. The Manchester carry adder and the lower-swing sense amplifier cut the read latency from about 270 ps to about 220 ps (about 18%) and reduced sense amplifier switching energy by 36%. The cell and array are unchanged by these optimizations, so the noise margins hold.

The main challenges were sizing the cell for both read stability and writability, developing enough bitline difference for the sense amplifier, and closing timing through the eight-stage decoder. Further work: a carry look-ahead adder, a domino decoder, write drivers with column multiplexing, and a proper corner and Monte Carlo study of the sense amplifier offset.